Graphing Techniques Topic Test

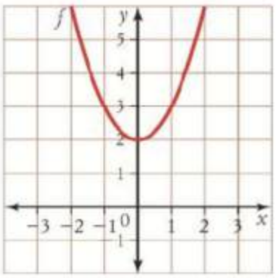
Name: ______________________

For Questions 1 – 3, circle the correct answer.

Question 1 1

Which graph shows $f(x) = x^2 + 2$?

A



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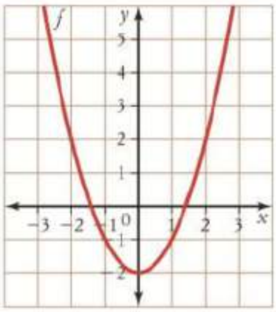

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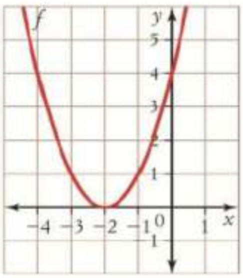

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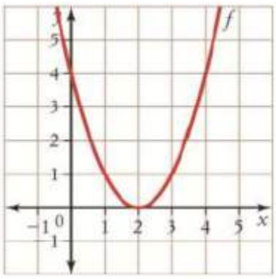
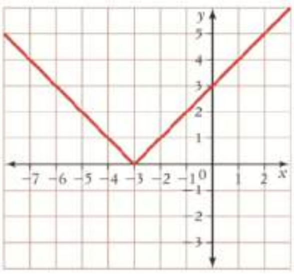


Question 2 1

Which graph shows $y = |x + 3|$?

A



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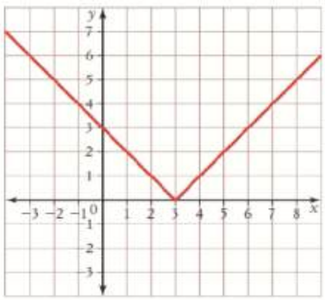

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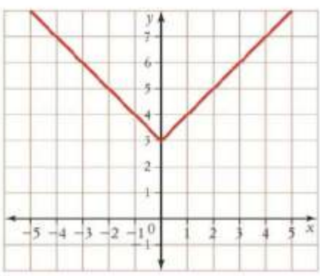

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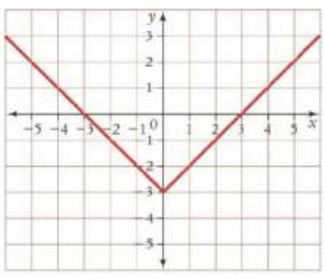
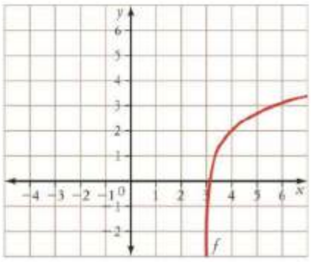


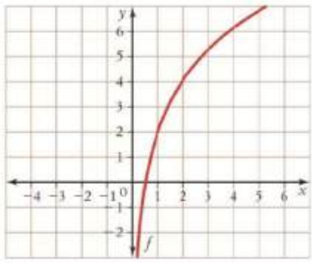
Question 3 1

Which graph shows $f(x) = 3lnx + 2$?

A



B



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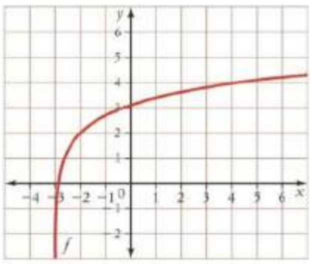

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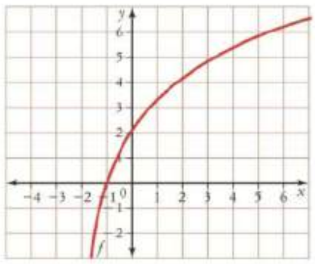

Question 4 3

Find the equation of the transformed function if $f(x) = x^2 - 7x + 2$ is:

a) Translated 2 units up.

____

____

b) Vertically dilated by a factor of 3.

____

____

c) Horizontally dilated by a factor of 4.

____

____

Question 5 3

Find the domain and range of the function $y = \frac{1}{x+3}$ and graph this function.

Question 6 4

Find the equation of each translated function.

a) $y = 2^x$ translated 2 units to the left.

b) $y = \frac{1}{x}$ translated 8 units up.

c) $f(x) = |2x| - 2$ translated 3 units to the right.

d) $f(x) = lnx$ translated 2 units down and 5 units to the right.

Question 7 3

Find the original point $P(x, y)$ on the function $y = f(x)$ if the coordinates of its image is $P'(-2, 3)$ when the function was:

a) Translated 2 units up and 3 units to the left.

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b) Dilated horizontally by a factor of 3 and translated down by 4 units.

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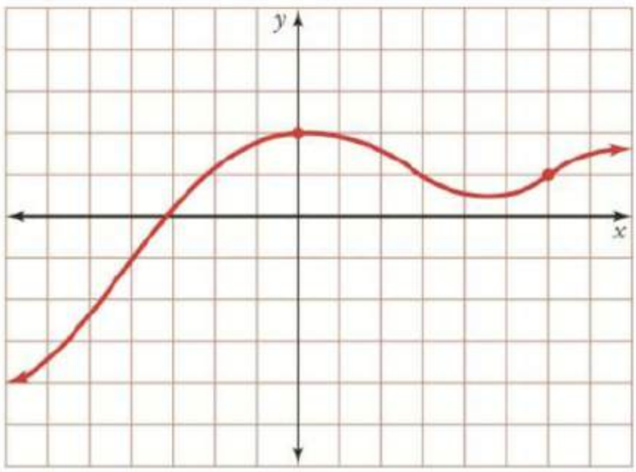
c) Translated right by 2 units and reflected along the x-axis.

Question 8 4

For the graph $y = f(x)$ shown, sketch a graph of the transformed functions:



a) $y = -2f(x - 1)$

b) $y = f(2x) + 3$

Question 9 4

Describe the transformations of $g(x) = x^3$, in order, that will give the transformed function $y = -3[4(x - 5)]^3 - 2$, stating the scale factors of both dilations.

Question 10 2

State the domain and range for the function $f(x) = 2e^{2x-4} + 3$?

Question 11 3

Find the coordinates of the point $P(x, y)$ on $y = f(x)$ if its image is $P'(-12, 8)$ when the function is transformed to:

a) $y = -2f(x + 3)$

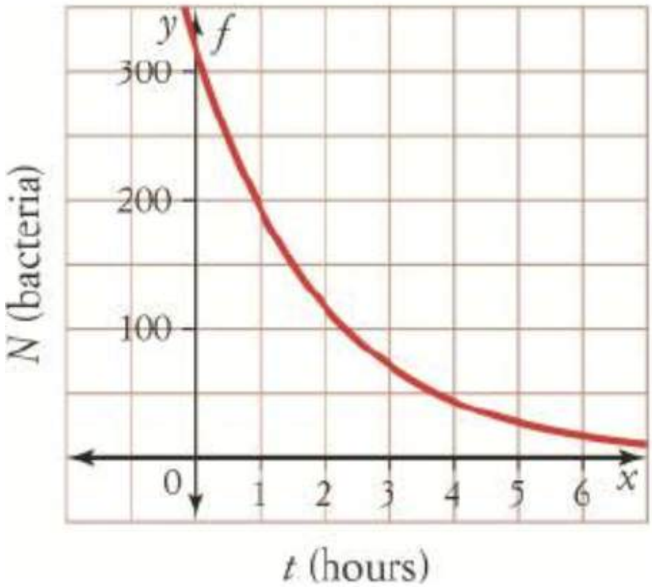
b) $y = 2f(-x) + 7$

c) $y = 3f(2x - 6) + 2$

Question 12

3

This graph of $N = 320e^{-0.5t}$ shows the number of bacteria, N, in a culture t hours after antibiotics are administered.



a) Use the graph to determine the time needed for the number of bacteria to halve.

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b) Show algebraically, the number of bacteria remaining after 6 hours.

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c) Identify the transformation represented by the 320, and explain what it means in the context of the question.

End of Topic Test

/32

Graphing Techniques Topic Test

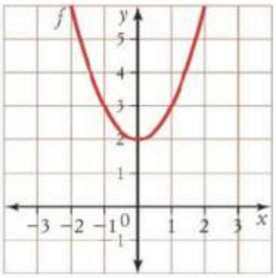
Name: ______________________

For Questions 1 – 3, circle the correct answer.

Question 1

1

Which graph shows $f(x) = x^2 + 2$?

A


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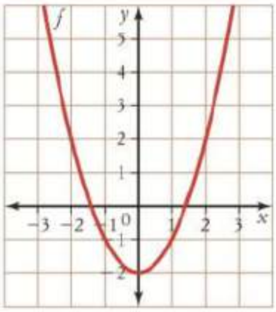

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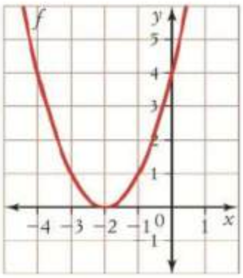
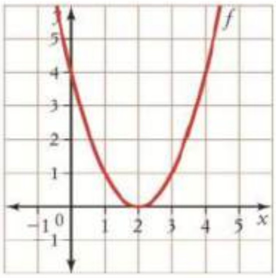
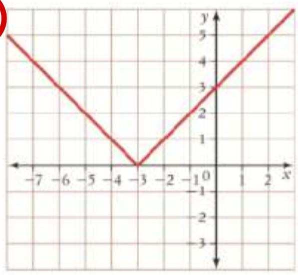


D


Question 2

1

Which graph shows $y = |x + 3|$?

A


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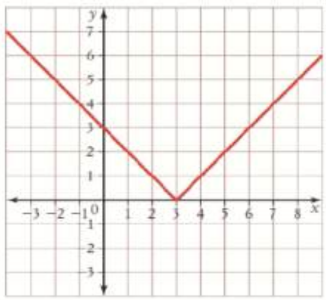

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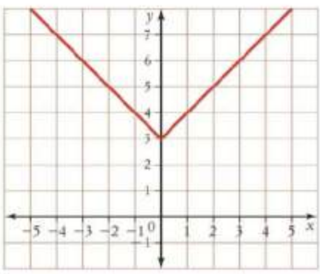

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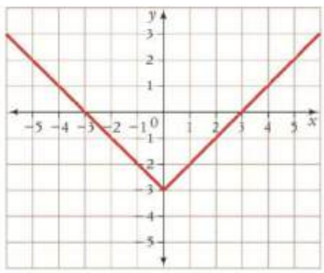
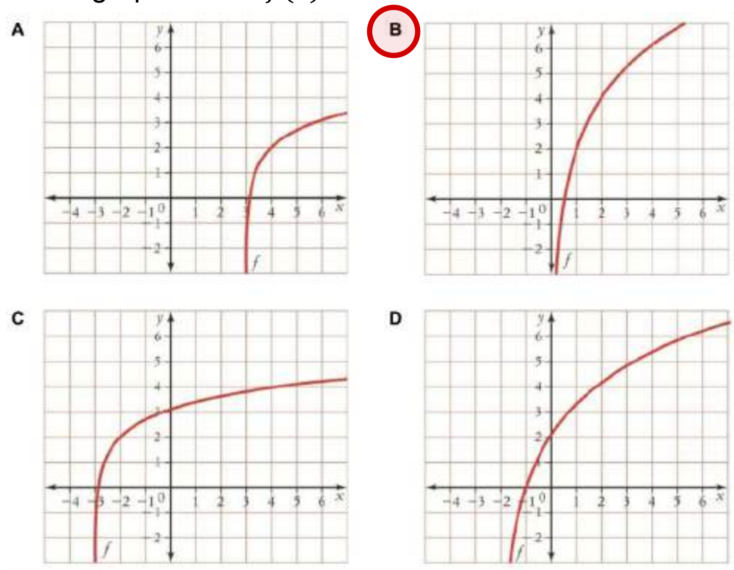


Question 3 1

Which graph shows $f(x) = 3lnx + 2$?



Question 4 3

Find the equation of the transformed function if $f(x) = x^2 - 7x + 2$ is:

a) Translated 2 units up.

$y = f(x) + 2$

$= x^2 - 7x + 4$

b) Vertically dilated by a factor of 3.

$y = 3f(x)$

$= 3x^2 - 21x + 6$

c) Horizontally dilated by a factor of 4.

$y = f\left(\frac{x}{4}\right)$

$= \left(\frac{x}{4}\right)^2 - 7\left(\frac{x}{4}\right) + 2 = \frac{x^2}{16} - \frac{7x}{4} + 2$

Question 5

3

Find the domain and range of the function $y = \frac{1}{x+3}$ and graph this function.

$y = \frac{1}{x}$ D: $(-\infty, 0) \cup (0, \infty)$

R: $(-\infty, 0) \cup (0, \infty)$

$y = \frac{1}{x+3}$ D: $(-\infty, 3) \cup (3, \infty)$

HT of 3 units left R: $(-\infty, 0) \cup (0, \infty)$

Question 6

4

Find the equation of each translated function.

a) $y = 2^x$ translated 2 units to the left.

$y = f(x+2)$

$= 2^{x+2}$

b) $y = \frac{1}{x}$ translated 8 units up.

$y = f(x) + 8$

$= \frac{1}{x} + 8$

c) $f(x) = |2x| - 2$ translated 3 units to the right.

$y = f(x-3)$

$= |2(x-3)| - 2$

d) $f(x) = lnx$ translated 2 units down and 5 units to the right.

$y = f(x-5) - 2$

$= \ln(x-5) - 2$

Question 7 3

Find the original point $P(x, y)$ on the function $y = f(x)$ if the coordinates of its image is $P'(-2, 3)$ when the function was:

a) Translated 2 units up and 3 units to the left.

$x - 3 = -2$ $\quad$ $y + 2 = 3$ $\quad$ $\therefore P(1, 1)$

$x = 1$ $\quad$ $y = 1$

b) Dilated horizontally by a factor of 3 and translated down by 4 units.

$3 \times x = -2$ $\quad$ $y - 4 = 3$ $\quad$ $\therefore P(-\frac{2}{3}, 7)$

$x = -\frac{2}{3}$ $\quad$ $y = 7$

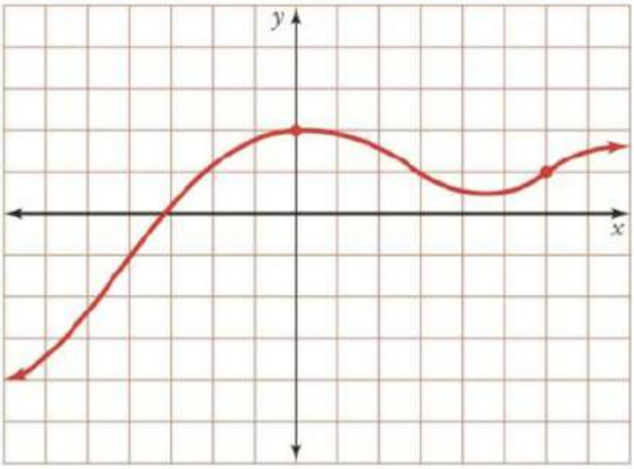
c) Translated right by 2 units and reflected along the x-axis.

$x + 2 = -2$ $\quad$ $y \times -1 = 3$ $\quad$ $\therefore P(-4, -3)$

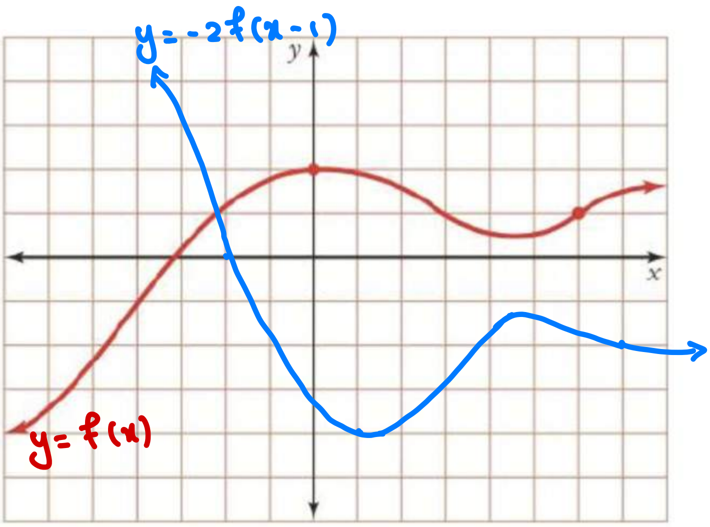
$x = -4$ $\quad$ $y = -3$

Question 8 4

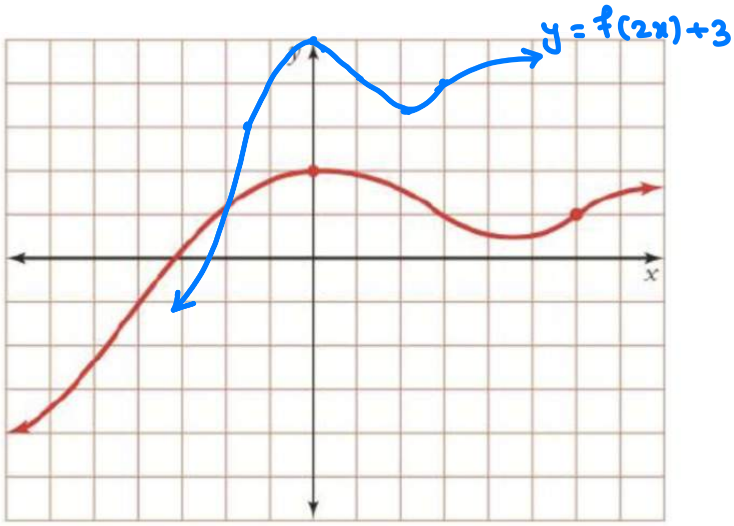
For the graph $y = f(x)$ shown, sketch a graph of the transformed functions:



a) $y = -2f(x - 1)$



b) $y = f(2x) + 3$



Question 9 4

Describe the transformations of $g(x) = x^3$, in order, that will give the transformed function $y = -3[4(x-5)]^3 - 2$, stating the scale factors of both dilations.

HD with SF $\frac{1}{4}$

HT of 5 units right

VD with SF -3

VT of 2 units down

Question 10 2

State the domain and range for the function $f(x) = 2e^{2x-4} + 3$?

Domain : $(-\infty, \infty)$

Range : $(3, \infty)$

Question 11 3

Find the coordinates of the point $P(x, y)$ on $y = f(x)$ if its image is $P'(-12, 8)$ when the function is transformed to:

a) $y = -2f(x+3)$

HT of 3 units left : $x - 3 = -12$ $\therefore x = -9$

VD of SF -2 : $y \times -2 = 8$ $\therefore y = -4$ $\therefore P(-9, -4)$

b) $y = 2f(-x) + 7$

Reflect in y-axis : $x \times -1 = -12$ $\therefore x = 12$ $\therefore P(12, \frac{1}{2})$

VD SF 2 & VT of 7 units up : $y \times 2 + 7 = 8$ $\therefore y = \frac{1}{2}$

c) $y = 3f(2x-6) + 2$ $= 3f(2(x-3)) + 2$

HD SF $\frac{1}{2}$: $x \times \frac{1}{2}$

HT of 3 units right : $x \times \frac{1}{2} + 3 = -12 \rightarrow \frac{x}{2} = -15$ $\therefore x = -30$

VD SF 3 : $y \times 3$

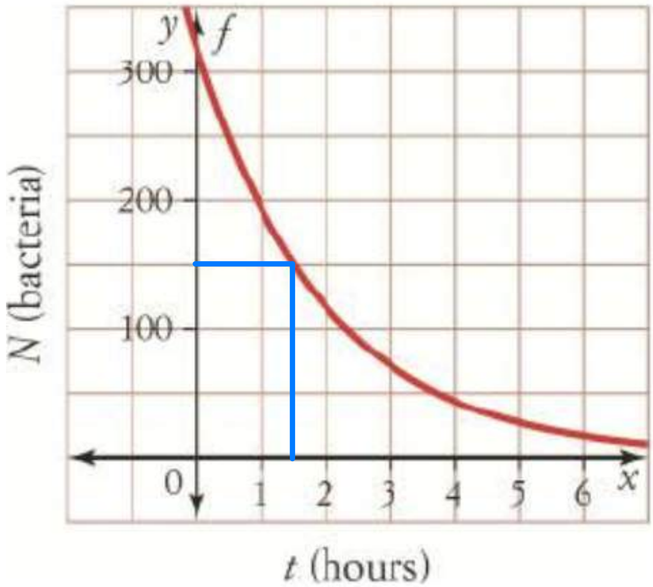
VT of 2 units up : $3y + 2 = 8 \rightarrow 3y = 6$ $\therefore y = 2$

$\therefore P(-30, 2)$

Question 12

3

This graph of $N = 320e^{-0.5t}$ shows the number of bacteria, N, in a culture t hours after antibiotics are administered.



a) Use the graph to determine the time needed for the number of bacteria to halve.

$t \approx 1.5$ hours

b) Show algebraically, the number of bacteria remaining after 6 hours.

Sub $t=6$, $N = 320e^{-0.5(6)} = 16$ bacteria

c) Identify the transformation represented by the 320, and explain what it means in the context of the question.

320 is a scale factor of a vertical dilation.
It is the initial number of bacteria in the culture.

End of Topic Test